

Element B: Documentation and Analysis of Prior Design Attempts

The problem at hand is the exclusion of students with disabilities, particularly those with learning disabilities, from participating in physical and social activities with their peers. This exclusion leads to feelings of isolation, low self-esteem, and hindrance in cognitive development among these students. Statistics reveal that approximately 7.3 million students in the US alone have learning disabilities, representing a significant portion of the population facing these challenges. Specifically, ultimate frisbee is a major sport that fosters intense physical activities which makes it challenging for special needs students to participate in. Addressing this issue is crucial for fostering inclusivity, promoting mental well-being, and improving overall academic outcomes for students with disabilities.

To throw the frisbee, or any object in general, an attempt was made to use a catapult to launch the frisbee a certain distance. The attempt addresses the needs of the special needs student as it allows for them to take part in not only the ultimate frisbee activity but many other activities associated with throwing objects. The catapult entails the following advantages: lightweight(allows for easy maneuverability), easy operation(needs minimal interaction to use), uses less parts(allows for easy repairability and replacement of parts). Despite these advantages, there are shortcomings which needs to be considered: limited performance-the small scale nature of the catapult does not allow for the frisbee to be launched far, unreliable performance-since the catapult uses small forces to throw the frisbee, over a short period of time, the performance of the frisbee will be minimized.

This attempt is specified toward baseball, a sport that involves the use of a bat and a ball that needs to be pitched. For practice, players often used an automatic pitcher which incorporated the use of high speed motors that spin flywheels that could launch the ball at varied speeds. The advantages that are entailed with this are: great performance-by optimizing the speed of the wheels that launch the object, various distances can be launched essentially making the special needs kids seem more approachable to other kids, long-term reliability-since the pitcher uses electricity, the student can operate it for longer periods of time. The drawbacks associated with this are: less maneuverability-since the parts needed to build the pitcher/launcher are required to maintain structural integrity, high quality and heavy parts need to be used, lack of repairability-an inexperienced teacher, in the case of a malfunction, will not be able to fix the problem or replace a part due to its complexity.

An attempt was made to modify a walker to facilitate participation in physical activities such as ultimate frisbee, offering versatility and familiarity for students with mobility impairments. While this modification addresses the need for inclusivity and promotes social interaction, it poses drawbacks including potential compromises to the walker's stability and functionality, particularly during intense physical activities. The advantages of these modifications are: ease of maneuverability-walkers are typically lightweight and portable, allowing for easy movement on different surfaces, facilitating participation in various environments, cost-effective-walkers are generally more affordable than specialized mobility devices, making modifications a cost-effective solution for promoting inclusivity in physical education programs. The drawbacks associated with modifications to the walker also need to be addressed: hindered repairability-unlike a traditional wheelchair, modifications to a walker needs to consider the user's personal interaction with the walker and create specialized parts to allow for optimal performance, lack of performance-despite specialization of parts, the limited mobility of the student using the walker will not foster a safe and effective method of launching the frisbee.